

413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 7

Prof. Suryanarayana Maddu
Department of Applied Mathematics and Statistics
Data Science and AI Institute
Johns Hopkins University

RECALL

Suppose we collect n independent and identically distributed observations

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p(x), \quad \mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2.$$

Recall that the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

is itself a random variable, with

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}, \quad \text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

By the Central Limit Theorem (CLT), for sufficiently large n ,

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0, 1).$$

In the previous lecture, we introduced the standard normal CDF

$$\Phi(z) = P(Z \leq z).$$

For example,

$$P(-1.96 \leq Z \leq 1.96) = 2\Phi(1.96) - 1 \approx 0.95.$$

Today, we use these results to quantify uncertainty in our estimates and evaluate scientific hypotheses.

1 Confidence Intervals

Suppose we observe a random sample $X_1, \dots, X_n$ from a population with unknown mean μ . For the samples, we can compute the sample mean $\bar{X}_n$.

The objective of confidence intervals is to construct an interval around the sample mean that contains the true population mean with a specified confidence level, such as 95%

1.1 Constructing a 95% Confidence Interval

From the CLT,

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0, 1).$$

Since approximately 95% of the probability under the standard normal distribution lies between -1.96 and

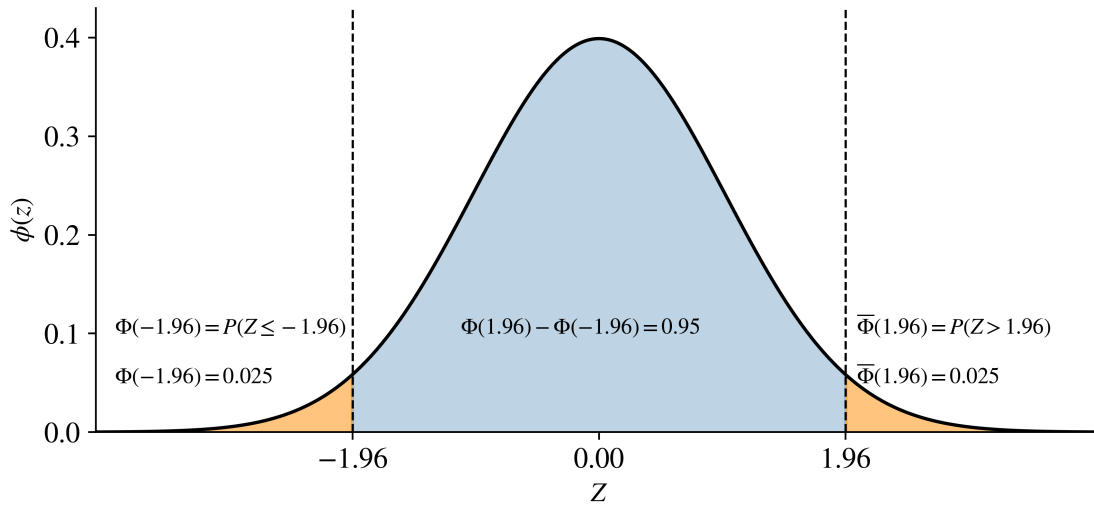


Figure 1: **Central 95% probability under the standard normal distribution.** For $Z \sim \mathcal{N}(0, 1)$, $P(-1.96 \leq Z \leq 1.96) = \Phi(1.96) - \Phi(-1.96) \approx 0.95$. The two tail regions each contain probability approximately 0.025, so that $P(Z \leq -1.96) \approx 0.025$ and $P(Z > 1.96) \approx 0.025$. This central probability is used to construct an approximate 95% confidence interval for a population mean.

1.96,

$$P\left(-1.96 \leq \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq 1.96\right) \approx 0.95.$$

Rearranging the inequalities gives

$$P\left(\bar{X}_n - 1.96 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + 1.96 \frac{\sigma}{\sqrt{n}}\right) \approx 0.95.$$

Thus, a 95% confidence interval for the population mean is

$$\text{CI}_{95\%} = \left[\bar{X}_n - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{X}_n + 1.96 \frac{\sigma}{\sqrt{n}} \right].$$

The quantity

$$\text{SE}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$$

is called the **standard error** of the sample mean.

What if σ is unknown? In practice, we estimate the population variance using

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Replacing σ with S_n gives the commonly used large-sample approximate confidence interval

$$\text{CI}_{95\%} \approx \bar{X}_n \pm 1.96 \frac{S_n}{\sqrt{n}}.$$

Here $S_n/\sqrt{n}$ is the estimated standard error. NOTE: For small samples from a Gaussian population with unknown σ , the appropriate reference distribution is Student's t -distribution rather than the standard normal distribution.

1.2 Interpreting a Confidence Interval

Example. Suppose we independently measure the waiting times of $n = 100$ patients at a hospital. The observed sample mean and standard deviation are

$$\bar{x}_n = 50 \text{ min}, \quad s_n = 20 \text{ min.}$$

The estimated standard error is

$$\widehat{\text{SE}}(\bar{X}_n) = \frac{s_n}{\sqrt{n}} = \frac{20}{\sqrt{100}} = 2 \text{ min.}$$

The approximate 95% confidence interval for the population mean waiting time is therefore

$$\begin{aligned} \text{CI}_{95\%} &= \bar{x}_n \pm 1.96 \frac{s_n}{\sqrt{n}} \\ &= 50 \pm 1.96(2) \\ &= [46.08, 53.92] \text{ min.} \end{aligned}$$

We estimate the population mean waiting time to be 50 minutes, with an approximate 95% confidence interval from 46.08 to 53.92 minutes.

IMPORTANT: What does 95% confidence mean? The population mean μ is a fixed but unknown parameter. The confidence interval is random because its endpoints depend on the collected sample.

If we repeatedly collected independent samples and constructed a confidence interval from each sample, approximately 95% of these intervals would contain the true population mean.

NOTE: Once an interval has been calculated from the observed data, its endpoints are fixed. We should not interpret the frequentist confidence level as a 95% probability that the fixed parameter μ lies within that particular observed interval.

2 Introduction to Hypothesis Testing

So far, we have used observed data to estimate unknown population parameters and quantify uncertainty in those estimates.

However, scientific experiments often begin with a different question:

Did an intervention produce a measurable change?

For example, we may want to investigate whether a new drug improves patient recovery, whether a vaccine reduces the risk of infection, or whether a new manufacturing process reduces the defect rate.

2.1 Control and Treatment

A common experimental design compares two groups:

- **Control group:** Receives the existing treatment, a placebo, or no intervention.
- **Treatment group:** Receives the new intervention whose effect we want to investigate.

Suppose p_C and p_T denote the probabilities of recovery in the control and treatment groups, respectively. We want to determine whether the observed difference between the groups reflects a genuine treatment effect or could arise from random sampling fluctuations (also called fluke in layman terms).

The null hypothesis (H_0). The null hypothesis represents a reference model against which we evaluate the observed data. In a treatment experiment, a natural choice for the null hypothesis is that the treatment has no effect:

$$H_0 : p_T = p_C.$$

The alternative hypothesis (H_1). The alternative describes the effect we want to investigate. For example, if the treatment is expected to increase recovery,

$$H_1 : p_T > p_C.$$

More generally, we may consider a two-sided alternative,

$$H_1 : p_T \neq p_C,$$

when changes in either direction are of interest.

Before-and-after experiments. Another common approach is to measure the same population before and after an intervention. If X_i^{before} and X_i^{after} are measurements on individual i , define

$$D_i = X_i^{\text{after}} - X_i^{\text{before}}.$$

A natural null hypothesis is

$$H_0 : \mathbb{E}[D_i] = 0.$$

However, observing a change after an intervention does not by itself establish that the intervention caused it. Other factors may change over time. An appropriately designed control group helps distinguish treatment effects from such alternative explanations.

2.2 Example: Did a New Teaching Method Improve Test Scores?

Suppose the historical mean score in a class is

$$\mu_0 = 65.$$

A new teaching method is introduced. We then observe the scores of $n = 25$ students and obtain a sample mean

$$\bar{X}_n = 70.$$

Suppose, for now, that the population standard deviation is known from historical data to be

$$\sigma = 10.$$

The observed mean has increased by 5 points. The question is:

Is this increase evidence that the population mean has genuinely increased, or could it simply be a random sampling fluctuation?

We organize a hypothesis test into three basic steps:

- Step 1:** Assume the null hypothesis H_0 is true,
Step 2: Construct a test statistic and determine its distribution under H_0 ,
Step 3: Determine how unusual the observed statistic is under H_0 .

Step 1: Assume the null hypothesis is true

The null hypothesis represents the reference model that “nothing has changed.” Thus,

$$H_0 : \mu = 65.$$

Since we are specifically interested in whether the new teaching method *increased* the mean score, the alternative hypothesis is

$$H_1 : \mu > 65.$$

Under H_0 , the true population mean remains 65. The observed value $\bar{x}_n = 70$ must therefore be interpreted as a possible random fluctuation of the sample mean around 65.

Step 2: Construct a test statistic

The observed deviation from the value predicted by H_0 is

$$\bar{X}_n - \mu_0 = 70 - 65 = 5.$$

However, the size of this deviation alone does not tell us whether it is unusual. We must compare it with the typical sampling fluctuation of the sample mean. Recall that for independent observations with variance σ^2 ,

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n},$$

so that

$$\text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

We therefore measure the deviation from H_0 in units of the standard deviation of the sample mean:

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}.$$

This is our **test statistic**. It measures how many standard errors the observed sample mean lies above or below the value predicted by H_0 .

What is the distribution of Z under H_0 ?

If the individual scores are independent Gaussian random variables,

$$X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2),$$

then the sample mean is exactly Gaussian for every n :

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Therefore, under $H_0 : \mu = \mu_0$,

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1).$$

If the individual scores are not Gaussian, this result is no longer exact. However, for sufficiently large n , the Central Limit Theorem gives

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0, 1),$$

so the same test can be used approximately.

For our observed data,

$$z_{\text{obs}} = \frac{70 - 65}{10/\sqrt{25}} = \frac{5}{2} = 2.5$$

Thus, the observed sample mean lies 2.5 standard errors above the mean predicted by H_0 .

Step 3: Determine how unusual the observation is under H_0

Under H_0 , we have established that

$$Z \sim \mathcal{N}(0, 1).$$

Thus, the null hypothesis provides a probability distribution for the values of the test statistic that we would expect to observe purely from random sampling. Values of Z near zero are common under H_0 , while values far from zero are increasingly unlikely. We therefore ask:

If H_0 were true, what is the probability of obtaining a test statistic at least as large as $z_{\text{obs}} = 2.5$?

This probability is the area under the standard normal density to the right of 2.5 (See Figure):

$$\begin{aligned} p &= P(Z \geq 2.5 \mid H_0) \\ &= 1 - \Phi(2.5) \\ &\approx 0.0062. \end{aligned}$$

This probability is called the **p-value**. It tells us how often a result at least this extreme would occur if the null hypothesis were actually true. Thus, if the true population mean were still 65, only about

$$0.62\%$$

of repeated samples would produce a sample mean at least this far above 65, when measured in units of its standard error.

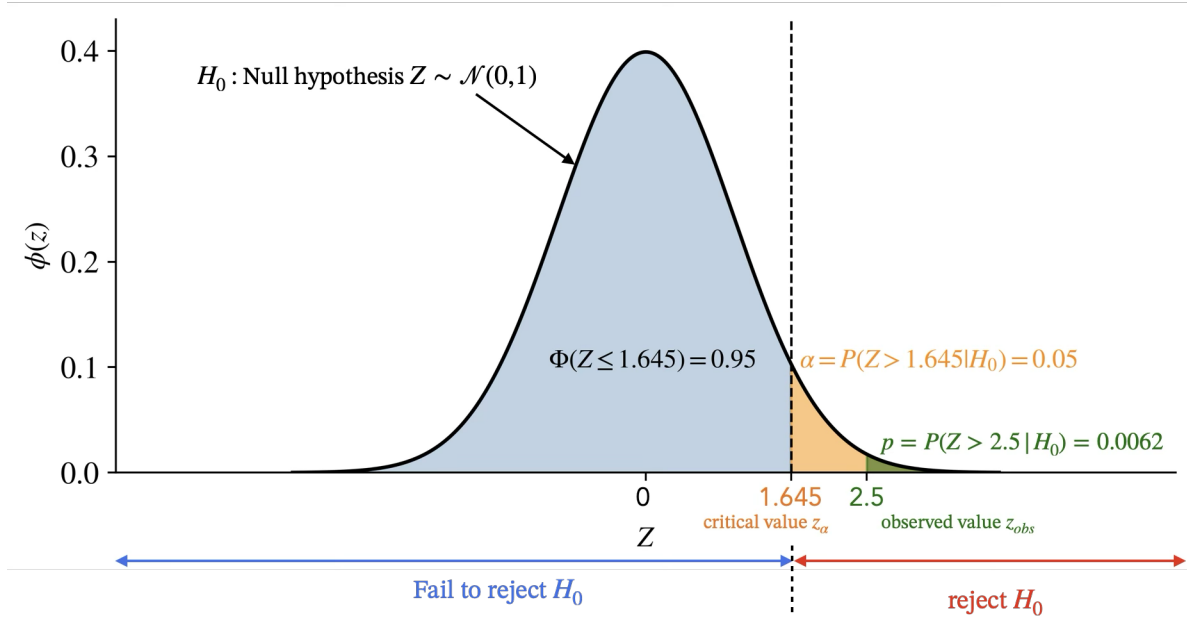


Figure 2: **One-sided hypothesis test under the null hypothesis.** Under H_0 , the test statistic follows the standard normal distribution, $Z \sim \mathcal{N}(0,1)$. For a significance level $\alpha = 0.05$, the critical value is $z_\alpha = 1.645$, so values $Z > 1.645$ fall in the rejection region. The observed statistic is $z_{\text{obs}} = 2.5$, giving the p-value $p = P(Z \geq 2.5 \mid H_0) \approx 0.0062$. Since $p < \alpha$, the observed result lies in the rejection region and we reject H_0 .

From a p-value to a decision

The p-value tells us how unusual the observed result is under H_0 . To turn this probability into a decision rule, we must decide *before looking at the data* how unlikely an observation must be before we are willing to reject H_0 .

We choose a small probability threshold α , called the **significance level**. A common choice is

$$\alpha = 0.05.$$

Choosing a significance level of $\alpha = 0.05$ means that if H_0 were actually true and we repeated the experiment many times, we would expect to incorrectly reject H_0 in about 5% of those experiments purely because of random sampling fluctuations.

For a one-sided test with

$$H_1 : \mu > \mu_0,$$

we therefore choose a cutoff z_α such that

$$P(Z \geq z_\alpha \mid H_0) = \alpha.$$

For $\alpha = 0.05$ and $Z \sim \mathcal{N}(0, 1)$,

$$z_\alpha \approx 1.645,$$

because

$$P(Z \geq 1.645) \approx 0.05.$$

Thus, our decision rule is

$$\begin{array}{ll} Z_{\text{obs}} \geq 1.645 & \implies \text{Reject } H_0, \\ Z_{\text{obs}} < 1.645 & \implies \text{Fail to reject } H_0. \end{array}$$

In our example,

$$z_{\text{obs}} = 2.5 > 1.645,$$

which lies well inside the rejection region. Therefore,

$$\text{Reject } H_0.$$

The same decision can be expressed using the p-value. Recall that

$$p = P(Z \geq 2.5 \mid H_0) \approx 0.0062.$$

Since

$$p = 0.0062 < \alpha = 0.05,$$

the observed result is more extreme than the threshold we agreed to regard as sufficiently unlikely under H_0 .

$$p \leq \alpha \iff \text{observation lies in the rejection region.}$$

Therefore, the observed result is sufficiently unusual under H_0 , and we

$$\text{Reject } H_0.$$

Conclusion. The observed sample mean of 70 is difficult to explain by ordinary sampling fluctuations around a population mean of 65. The data therefore provide statistical evidence that the population mean score has increased.

What assumptions did we make?

For the calculation above, we assumed:

1. The observations $X_1, \dots, X_n$ are independent.
2. They are sampled from the same population with mean μ and variance σ^2 .
3. The population standard deviation $\sigma = 10$ is known.
4. Either the population distribution is Gaussian, in which case

$$Z \sim \mathcal{N}(0, 1)$$

exactly, or the sample size is sufficiently large for the Central Limit Theorem to provide a good approximation.

If the population standard deviation σ is *unknown*, we must estimate it from the observed data. The resulting test statistic has a different reference distribution for finite samples; we will return to this case later.

2.3 Why Does Sample Size Matter?

Suppose we observe the same difference in means,

$$\bar{X}_n - \mu_0 = 70 - 65 = 5,$$

and the same standard deviation $\sigma = 10$, but change the number of observations n . The test statistic is

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} = \frac{5}{10/\sqrt{n}}.$$

Therefore, the strength of the evidence depends not only on the observed difference, but also on the sample size.

n	SE $\sigma/\sqrt{n}$	z_{obs}	One-sided p-value
9	3.33	1.50	≈ 0.0668
25	2.00	2.50	≈ 0.0062
100	1.00	5.00	$\approx 2.9 \times 10^{-7}$

For a small sample, the sample mean fluctuates substantially from one sample to another. Therefore, a difference of 5 points may still be reasonably explained by sampling variability. For example, when $n = 9$,

$$p \approx 0.0668 > 0.05,$$

so we fail to reject the null hypothesis at the 5% significance level.

For a larger sample size n , the standard error

$$\frac{\sigma}{\sqrt{n}}$$

becomes smaller. The same 5-point difference is then much harder to explain as a statistical fluctuation under H_0 .

$n \uparrow \implies \text{SE}(\bar{X}_n) \downarrow \implies Z \uparrow \implies \text{stronger evidence against } H_0$
--

Key point: Statistical significance depends on the size of the observed effect *relative to its sampling uncertainty*, not simply on the raw difference between two means.

2.4 Example: Is a Coin Biased Toward Heads?

Suppose we flip a coin $n = 20$ times and observe

$$H_{\text{obs}} = 14$$

heads. We want to test whether the coin is biased toward heads.

Step 1: Formulate the hypotheses:

$$H_0 : p = 0.5, \quad H_1 : p > 0.5.$$

Under H_0 , each flip is Bernoulli with success probability 0.5. Therefore, the total number of heads

$$H = \sum_{i=1}^{20} X_i$$

has the exact distribution

$H \sim \text{Binomial}(20, 0.5)$	under H_0 .
-----------------------------------	---------------

Step 2: Choose the test statistic: We use

$$T = H,$$

the total number of heads. Since the alternative hypothesis is $H_1 : p > 0.5$, unusually large values of H provide evidence against H_0 .

Step 3: Determine how unusual the observation is under H_0 : We observed

$$H_{\text{obs}} = 14.$$

The one-sided p-value is therefore

$$\begin{aligned} p &= P(H \geq 14 \mid H_0) \\ &= \sum_{k=14}^{20} \binom{20}{k} \left(\frac{1}{2}\right)^{20}. \end{aligned}$$

Evaluating the sum,

$$p \approx 0.0577.$$

Thus, if the coin were actually fair, there would be about a 5.77% probability of observing 14 or more heads in 20 flips. At significance level $\alpha = 0.05$, we have

$$p = 0.0577 > 0.05.$$

Therefore,

Fail to reject H_0 .

The data do not provide sufficient evidence, at the 5% significance level, to conclude that the coin is biased toward heads.

2.5 Example: Is a Filling Machine Underfilling Bottles?

Suppose a filling machine is designed to dispense 100 mL into each bottle. Previous calibration established that the population standard deviation is $\sigma = 4$ mL.

An independent random sample of $n = 64$ bottles has observed mean $\bar{x}_n = 99$ mL. We want to determine whether the machine is systematically underfilling bottles.

Step 1: Formulate the hypotheses.

Let μ denote the population mean filling volume. We test

$$H_0 : \mu = 100 \text{ mL}, \quad H_1 : \mu < 100 \text{ mL}.$$

Under H_0 , the machine is filling bottles correctly on average. The observed mean $\bar{x}_n = 99$ mL must therefore be explained by random sampling fluctuations.

Step 2: Choose the test statistic.

We use the sample mean $\bar{X}_n$. Under H_0 ,

$$\mathbb{E}[\bar{X}_n] = 100,$$

and

$$\text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{64}} = 0.5 \text{ mL}.$$

We standardize the deviation of the sample mean from the value predicted by H_0 :

$$Z = \frac{\bar{X}_n - 100}{4/\sqrt{64}}.$$

Using the CLT, or exactly if the individual filling volumes are Gaussian,

$$\boxed{Z \approx \mathcal{N}(0, 1) \quad \text{under } H_0.}$$

For the observed sample,

$$z_{\text{obs}} = \frac{99 - 100}{0.5} = -2.$$

Thus, the observed sample mean lies two standard errors below the mean predicted by H_0 .

Step 3: Determine how unusual the observation is under H_0 .

Since the alternative hypothesis is

$$H_1 : \mu < 100,$$

unusually small values of Z provide evidence against H_0 . The one-sided p-value is therefore

$$\begin{aligned} p &= P(Z \leq z_{\text{obs}} \mid H_0) \\ &= P(Z \leq -2) \\ &= \Phi(-2) \\ &\approx 0.0228. \end{aligned}$$

If the true mean filling volume were 100 mL, there would be about a 2.28% probability of observing a sample mean at least this far below 100 mL due to random sampling fluctuations. At significance level $\alpha = 0.05$,

$$p = 0.0228 < 0.05,$$

so we

Reject H_0 .

The sample provides statistical evidence that the *population mean* filling volume is below 100 mL. This does not imply that every individual bottle is underfilled.

What changes if $n = 16$ but $\bar{x}_n = 99$ mL and $\sigma = 4$ mL remain the same? Then

$$z_{\text{obs}} = \frac{99 - 100}{4/\sqrt{16}} = -1,$$

and the one-sided p-value becomes

$$p_{\text{value}} = \Phi(-1) \approx 0.159.$$

In this case, we fail to reject H_0 at the 5% significance level.

2.6 Example: Has a Chip Defect Rate Increased?

A semiconductor process historically produces defective chips with probability

$$p_0 = 0.05.$$

In a new production run, $n = 400$ chips are sampled independently and 30 are defective, so

$$\hat{p} = \frac{30}{400} = 0.075.$$

Step 1: Formulate the hypotheses.

$$H_0 : p = 0.05, \quad H_1 : p > 0.05.$$

Step 2: Construct the test statistic: For Bernoulli observations,

$$\text{SD}(\hat{p}) = \sqrt{\frac{p_0(1 - p_0)}{n}}.$$

Thus,

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} \approx \mathcal{N}(0, 1)$$

by the CLT. For the observed data,

$$z_{\text{obs}} = \frac{0.075 - 0.05}{\sqrt{0.05(0.95)/400}} \approx \boxed{2.29}.$$

Step 3: Compute the p-value: Since $H_1 : p > 0.05$, we use the right tail:

$$\begin{aligned} p &= P(Z \geq 2.29 \mid H_0) \\ &= 1 - \Phi(2.29) \\ &\approx 0.011. \end{aligned}$$

Since

$$p = 0.011 < 0.05,$$

we

Reject H_0 .

The data provide statistical evidence that the defect probability has increased above 5%.

Why is the normal approximation reasonable? Under H_0 ,

$$np_0 = 20, \quad n(1 - p_0) = 380,$$

so both expected counts are sufficiently large for the CLT approximation.